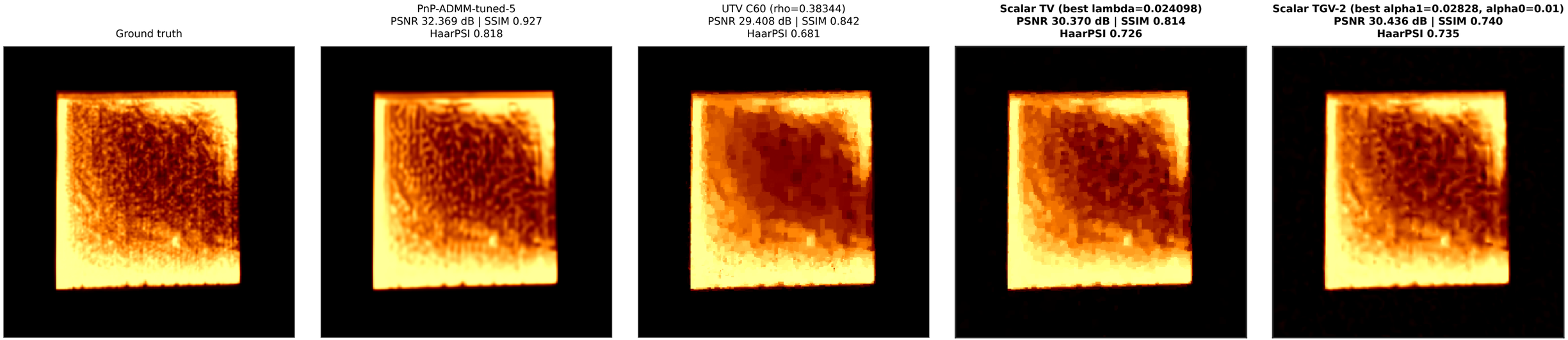
Perovskite 0292 | Best scalar TV and TGV-2 compared with PnP-ADMM and C60

Frozen 20% multilevel Hadamard measurement | NPL-default noise | all methods share the identical observation

TV, TGV-2 and C60 scalings are tuned for this image using ground-truth PSNR (oracle comparison)



Scalar TV: lambda = 0.02409825, 8,192 PDHG iterations, MRPro-backed componentwise TV.

Scalar TGV-2: alpha1 = 0.02828427, alpha0 = 0.01, 8,192 PDHG iterations, MRPro-backed componentwise TGV.

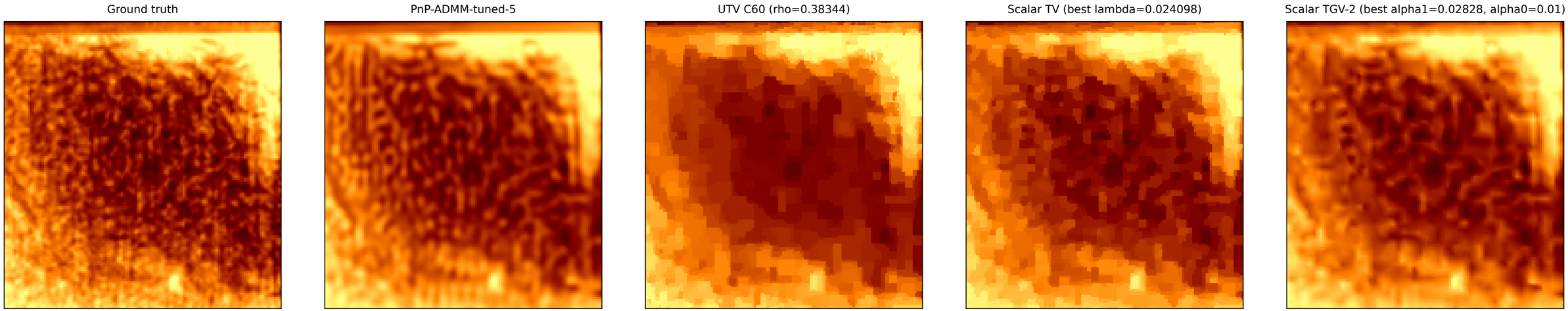
UTV C60: rho = 0.3834424, 512 PDHG iterations. PnP-ADMM: frozen tuned-5 parameters.

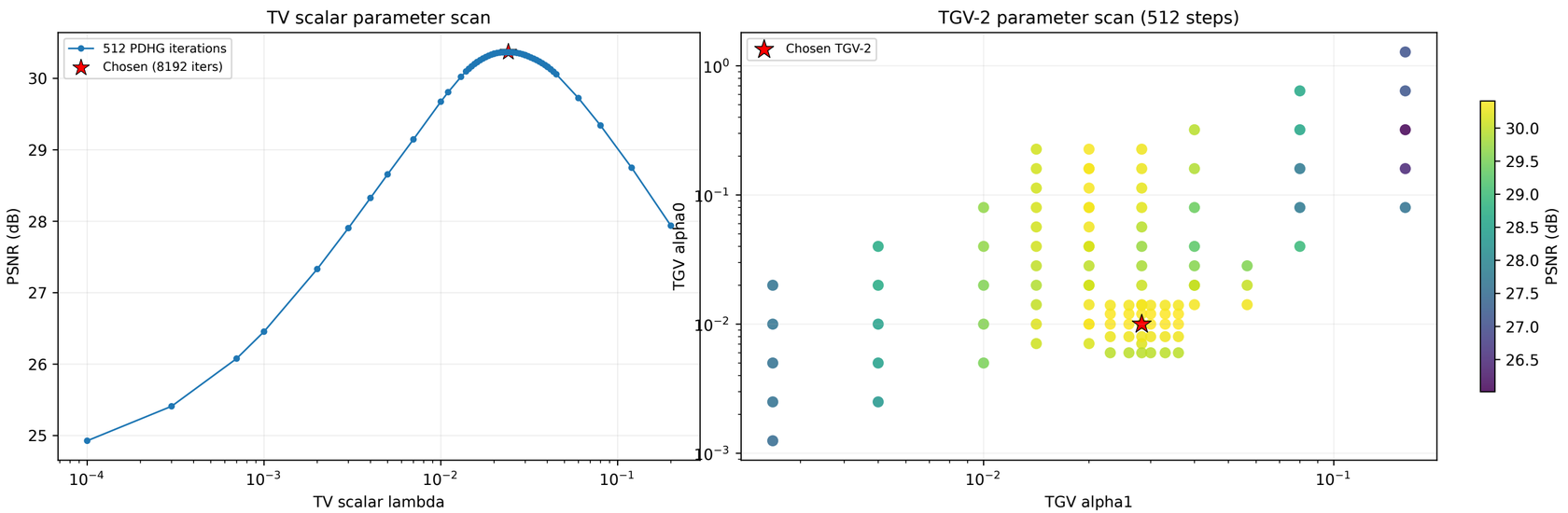
All PSNR/SSIM/HaarPSI metrics use raw (unclipped) images. Colour display fixed to [0,1].



Perovskite 0292 | Texture crop and scalar TV / TGV-2 tuning

Crop: rows 40:185, columns 65:205 | identical crop and [0,1] display scale in all methods





Selected scalar regularisation
TV lambda = 0.02409825
TGV-2 alpha1 = 0.02828427
TGV-2 alpha0 = 0.01

TV PSNR: 30.370 dB
TGV-2 PSNR: 30.436 dB
TV - TGV: -0.067 dB

TV vs PnP: -2.000 dB
TV vs C60: +0.961 dB

TV and TGV are both GT-tuned
for this single image.

Archived validation-tuned TV (lambda=0.011; 1024 iterations): PSNR 29.786 dB | not the oracle scalar TV selected for this comparison.

TGV-2 and TV implementations: existing LION AdaptiveTGVPCMNetwork / AdaptiveTVPCMNetwork with MRPro-backed PDHG. Scores and plots refer to the same frozen PCM measurement.